

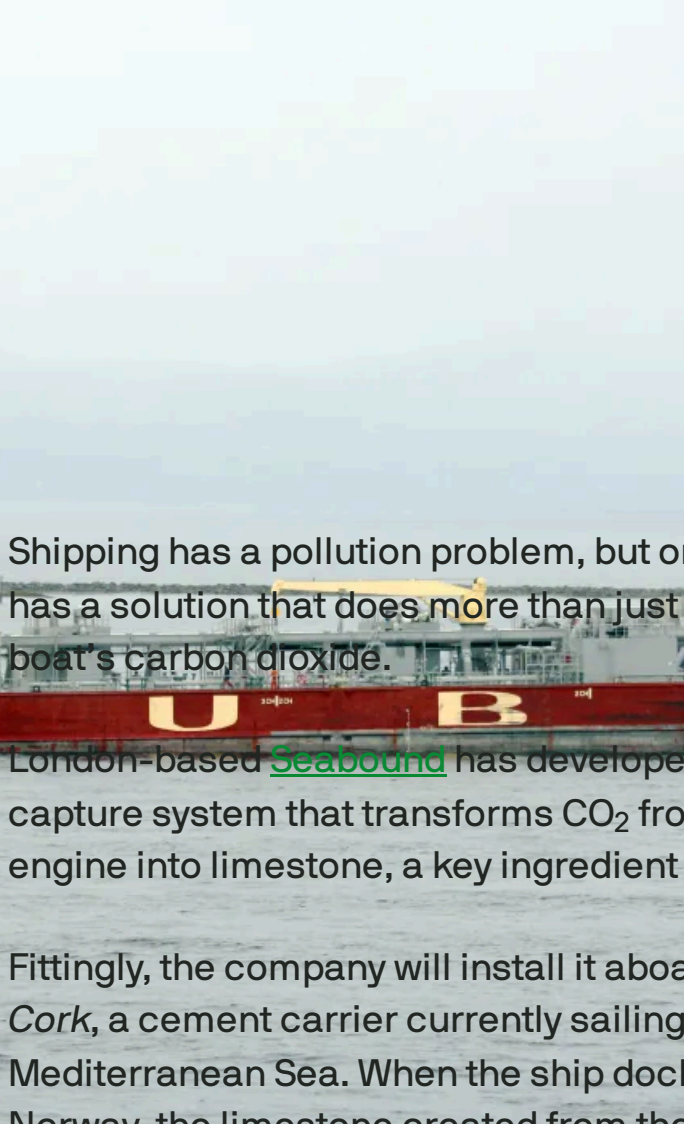
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CLIMATE



Meet the cement  
transport ship that  
makes cement  
ingredients while sailing



Shipping has a pollution problem, but one company has a solution that does more than just eliminate a boat's carbon dioxide.

London-based [Seabound](#) has developed a carbon capture system that transforms CO<sub>2</sub> from the engine into limestone, a key ingredient in cement.

Fittingly, the company will install it aboard the *UBC Cork*, a cement carrier currently sailing through the Mediterranean Sea. When the ship docks in Norway, the limestone created from the voyage will be offloaded and used to make more cement at [Heidelberg Materials](#)' net-zero plant in Brevik. (The name Heidelberg may ring a bell — earlier this year, it inked a deal to deploy more than 100 autonomous trucks from former Google exec [Anthony Levandowski's startup Pronto](#).)

Both maritime shipping and cement are highly polluting industries, representing about 3% and 8% of global carbon emissions, respectively.

Their emissions are challenging to address, too. For shipping, batteries are not currently energy-dense enough to enable the sorts of voyages many vessels undertake. And the chemical reaction that forms Portland cement, the most widely used type, releases carbon dioxide, to say nothing of the fossil fuels that typically drive the process.

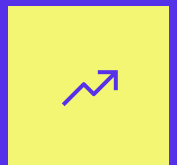
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There's some urgency for maritime shipping to rein in its pollution: The International Maritime Organization (IMO), which regulates the global shipping industry, will require owners [to trim greenhouse gas emissions](#) from their fleets by 30% over the next decade, rising to 65% by 2040.

Seabound is just one company developing potential solutions. Another, [Amogy](#), is [proposing](#) using its clever ammonia-cracking technology to deliver zero-emission power.

While ammonia has gained currency in the shipping industry as an energy-dense fuel with the potential to eliminate greenhouse gas emissions, its use would require ships to overhaul or completely replace their power plants.

Seabound is proposing a retrofit that would leave existing internal combustion engines intact, adding a carbon capture system that would tap into their exhaust pipes. Heidelberg Materials said that the use of Seabound's technology would help it reduce the emissions that result from shipping its cement.

Topics: [carbon capture](#) [cement](#) [Climate](#)

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**Tim De Chant**

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IMAGE CREDITS: CARBON CLEAN

Harri Weber — 11:01 AM PDT · May 11, 2022

[Carbon Clean](#), a startup focused on capturing harmful factory emissions before they enter the atmosphere, has secured a hefty \$150 million series C round led by oil company Chevron.

Aiming to carry out “industrial decarbonization on a gigatonne scale by the mid-2030s,” Carbon Clean is chasing a sector that is responsible for around one quarter of global greenhouse gas emissions — though the statistic [depends](#) on [how](#) you [define](#) “heavy industry,” which includes cement, steel, energy from waste and oil refineries.

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Carbon Clean claims its latest tech — dubbed the CycloneCC — is “10 times smaller” than conventional point-of-source carbon capture equipment and “has the potential” to cut [average costs](#) roughly in half, to around \$30 per tonne. Small enough to fit inside a shipping container, the CycloneCC takes in flue gases, absorbs the CO<sub>2</sub> via liquid chemicals known as [amine scrubbers](#), and then [boils off the CO<sub>2</sub>](#) for storage, say, [underground](#).

Asked about Chevron’s [reputation on climate](#), Carbon Clean co-founder and CEO Aniruddha Sharma employed a firefighting analogy and added that fossil fuel firms have more in-house expertise on emissions than your typical steel or cement plant.

“I say: Look, I’m a fireman and there is a fire. And if I have to start thinking about decarbonization as trying to douse the fire, I’ll go to the biggest fire first, because that needs to be controlled first,” he said. “These companies have such a big emissions footprint today already that, you know, once you start working there it can already make a big impact.”

In addition to Chevron, investors including cement producer CEMEX, oil and gas giant Saudi Aramco and Samsung Ventures also chipped in on the deal.

“The table is huge and there are just not enough companies out there doing this kind of thing,” Sharma said in a call with TechCrunch, arguing that point-of-source carbon capture capacity needs to expand by as much as 40 times in the next few years. “Really, when I think about competition, my competition is climate change. The clock is ticking,” he said.

Topics: [carbon capture](#) [Chevron](#) [Climate](#)

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Harri Weber was a senior writer at TechCrunch covering climate. Her work has also appeared in Gizmodo, Fast...

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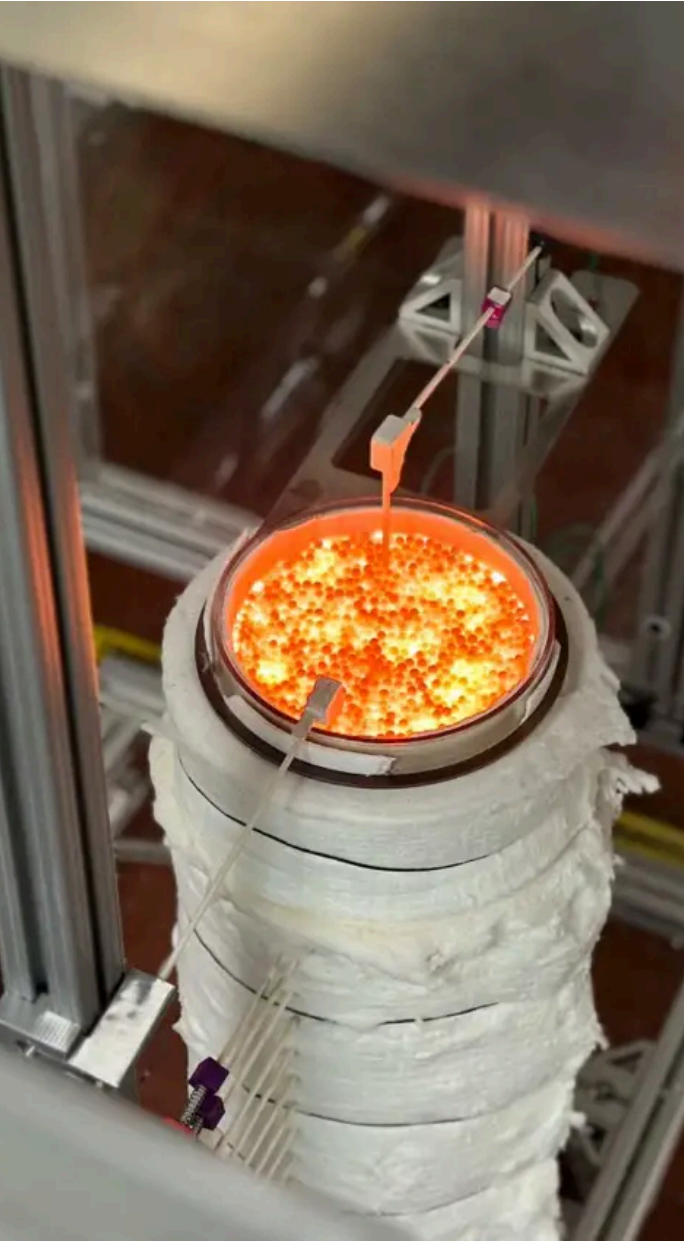


IMAGE CREDITS: FURNO

CLIMATE



# Furno bets that super-efficient, modular kilns will turn the cement industry upside down

Tim De Chant — 6:31 AM PDT · March 13, 2024



For every 1,000 kilograms of cement the world produces, 600 kilograms of carbon dioxide are dumped into the atmosphere. If that doesn't sound like a lot, consider that last year the world made about [4.2 billion metric tons](#) of cement, an amount so large that it's responsible for about 8% of all carbon pollution.

It's a big problem, one which weighed on Gurinder Nagra's mind as a graduate student at Stanford University. It wasn't the focus of his PhD — that would be carbon capture and storage — but he couldn't stop thinking about it. When he took a class on energy ventures three different times in two years, "every time was focused on cement," he told TechCrunch.

It wasn't just the pollution that bothered Nagra. "Emissions are one aspect. The other is that it's just this overlooked, dirty industry, but it's kind of necessary to modern civilization. It's everywhere," he said. Doing something in cement could have ripple effects far beyond climate change.

"The companies that control the supply of cement in the background run the world almost. Not so much as the oil industry, but they have a lot of power," he said. "It costs a billion dollars to build a new plant. If a company decides to build a plant in a developing economy or a developing country, that company has control over the destiny of the trajectory of that developing economy."

So, after finishing his degree in 2020, Nagra founded [Furno](#). The goal was straightforward but ambitious: upend the cement industry.

Many developing countries aren't able to finance a new plant or entice a company to build one, so they end up importing cement, paying dearly in the process. "You find that, in a lot of developing economies, they actually pay more for cement than

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### CLIMATE



**This Detroit startup is turning to utilities to make home efficiency upgrades cheaper**

Tim De Chant - Aug 1, 2025

in developed economies, sometimes three to four times the market price.”

Furno’s answer has been to develop a cement kiln that’s smaller, more efficient and more flexible than the massive ones favored by industry today. Each unit may produce less than today’s industry-standard designs, but it promises to be cheaper to build and use less energy to run.

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Furno's smaller kilns fly in the face of trends in the cement industry today, which has been chasing ever-larger equipment. Large rotary kilns can squeeze operating costs, though in the process, new plants have grown so large that each one is essentially a bespoke installation that incurs large labor costs to build. By building smaller kilns, Furno is hoping to lower capital expenditures through mass production and lower operating expenses through energy efficiency.

Here's how: Portland cement, the most commonly used type, is produced by heating calcium containing minerals, usually from limestone. Those minerals are sent through the kiln, where they're heated to more than 1,110 degrees F (600 degrees C), almost always using fossil fuels. The process of calcination, as it's known, produces lime (calcium oxide) and carbon dioxide. There are other steps that occur at a modern cement plant, but that's the most important one. Raw materials enter one end of the kiln, which lies horizontally, and are pushed past an intense flame to the other end, where they emerge as clinker.

Kilns today are not very efficient. A [study](#) of one in China showed that only about 30% of the heat in the kiln powered the calcination reaction; the rest was lost. "If you're standing 40 feet away from a kiln, you can feel a heat dissipating out on the walls of the unit," Nagra said.

Furno's kiln, on the other hand, is far more efficient, reducing emissions from fossil fuels by 70% or more. It accomplishes that by turning the kiln vertical, filling it with ingredients that are ground to prescribed dimensions, and then seeping through it a combustible gas (natural gas, biogas or hydrogen) and setting it ablaze. By varying the amount of gas and oxygen flowing through the column, Furno is able to control where the flame occurs inside, ensuring that more of the heat powers the

calcination process instead of escaping the kiln. Less fuel means less pollution, and carbon capture can help reduce emissions further. Raw material continuously flows into the top of the kiln, and clinker drops out the bottom.

What's more, because the flame front inside the kiln can be more precisely controlled, the company has a path toward using recycled cement from old concrete, Nagra said. Because the recycled cement has already been calcined, the kiln wouldn't need to use as much fuel. And because Furno's kilns are smaller, they could allow cement to be remade right on a construction site.

That approach has helped Furno raise a \$6.5 million seed round, TechCrunch has exclusively learned. The round was led by Energy Capital Ventures with participation by Cantos, Neotribe and O'Shaughnessy Ventures.

Already, Furno has captured the attention of ready-mix concrete companies, which have to buy cement from vertically integrated companies. "They have to buy cement from their competition, which doesn't really make for a very fair market." Modular kilns offer them the chance to gain some control over their own fates.

Furno's focus on kilns and the software that controls them sets them apart from many of their competitors, which instead focus on changing the chemical reaction itself. By improving the kiln and leaving the rest largely alone, Furno might find it easier to sell to an industry that's not accustomed to change.

Topics: [cement](#) [Climate](#) [construction](#)

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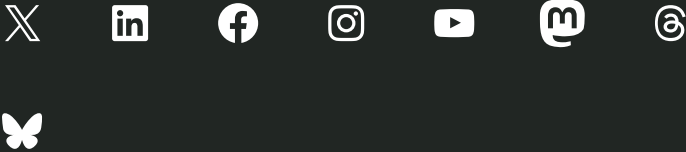
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